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MyMark: A Decentralised Media Fingerprinting Framework Integrating Invisible
Watermarking, Perceptual Hashing, Blockchain Provenance, and 4D Biometric Verification

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MyMark: A Decentralised Media Fingerprinting Framework Integrating Invisible Watermarking, Perceptual Hashing, Blockchain Provenance, and 4D Biometric Verification

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1.1 Abstract—

The protection of digital media and the safeguarding of online identities have become pressing challenges in an era where content can be replicated, manipulated, and redistributed at scale. Conventional copyright enforcement and age-gating measures have proven inadequate. Platforms such as Yubo continue to struggle to provide safe online environments for young people, while governments in the UK, Australia, and elsewhere are pursuing legislation that normalises age verification. As critics warn, this approach increases the transmission of sensitive personally identifiable information (PII) and undermines the principle of data minimisation, thereby exposing users to heightened risks of surveillance, tracking, and fraud (Politico, 2024).

This dissertation presents MyMark, a decentralised media fingerprinting system that integrates robust invisible watermarking, perceptual hashing, blockchain provenance, and 4D biometric verification. MyMark not only enables creators to embed and register ownership claims in a tamper-evident ledger, but also introduces a privacy-preserving mechanism for identity binding and liveness verification that does not require the storage of images or raw biometric data. In doing so, the system addresses both the intellectual property protection gap and the failures of existing age and identity verification regimes.

The contributions of this work include:

- (1) a hybrid detection pipeline combining watermarking and perceptual hashing;
- (2) a blockchain provenance layer storing cryptographic commitments only;
- (3) a biometric module capable of verifying age and liveness without retaining PII;
- (4) an evaluation framework measuring robustness, imperceptibility, accuracy, and ledger performance;
- (5) a critical analysis of the ethical and legal implications under GDPR and broader information security concerns.

Index Terms — Digital watermarking; perceptual hashing; blockchain provenance; media fingerprinting; 4D face recognition; biometric verification; intellectual property protection; age verification; privacy-preserving authentication; GDPR compliance.

1.2 Background and Motivation

The shift from analogue to digital media has radically altered the dynamics of content ownership and distribution. Unlike physical artefacts, digital objects can be replicated without degradation, making unauthorised use of intellectual property widespread and difficult to control (Lessig, 2008). Although watermarking and reverse image search tools exist, most are either easily defeated through common transformations (cropping, compression, filtering) or accessible only to large corporations, leaving individual creators under-protected (Cox et al., 2008; Fridrich, 2010).

At the same time, blockchain technology has introduced new possibilities for tamper-evident provenance tracking. Blockchains provide immutable records of ownership claims, but by themselves cannot bind content to individuals without additional mechanisms (Narayanan et al., 2016). Furthermore, the misuse of these systems—such as fraudulent ownership claims—remains a challenge (Zhao et al., 2020). Biometric technologies offer a solution to this issue by ensuring that only legitimate owners can register content, but concerns around spoofing and identity theft highlight the need for advanced techniques such as 3D/4D face recognition and liveness detection (Bowyer et al., 2020; Liu et al., 2022).

Beyond intellectual property, this project is also motivated by wider challenges in network and information security, particularly the handling of personal data in digital identity systems. Age-gating has become a central strategy for online safety, especially on youth-focused platforms such as Yubo, yet these measures have repeatedly proven ineffective at creating genuinely safe environments (Politico, 2024). Governmental attempts in the UK, Australia, and beyond to legislate age verification risk further

undermining privacy. As Politico (2024) observes: “The normalization of age verification, and the subsequent huge increase in the amount of identifiable data being transmitted online, contradicts the principle of data minimization and supports a growing trend of de-anonymization, leaving us all more vulnerable to tracking, surveillance, and instances of fraud.”

In this context, the central challenge is not simply whether age verification or identity-binding can be enforced, but whether it can be achieved without exposing individuals to new vectors of surveillance and exploitation. Current approaches demand extensive personally identifiable information (PII)—such as passport scans, facial photographs, or government databases—placing users, particularly young people, at risk of misuse and secondary harms (Livingstone et al., 2023).

This dissertation therefore presents a different approach: a privacy-preserving system that never stores images or raw biometric data, yet is still capable of verifying liveness, binding ownership, and supporting age-related assurances. By embedding data minimisation principles into its architecture, MyMark demonstrates that security and safety can be achieved without amplifying risks of surveillance, fraud, or identity theft. Crucially, this contribution moves beyond theoretical critique into implementation, providing a working artefact that integrates robust watermarking, decentralised provenance, and biometric verification without violating user privacy.

1.3 Research Problem

The core research problem addressed in this dissertation is:

How can individual creators be provided with an accessible, robust, and secure system for registering, proving, and defending ownership of digital media in a decentralised environment while preventing fraudulent claims?

This problem is multi-dimensional, involving:

Robustness: ensuring watermarks and fingerprints survive common transformations.

Accuracy: minimising false positives in perceptual matching.

Trustworthiness: ensuring provenance data cannot be altered.

Authenticity: preventing malicious users from registering content they do not own.

Usability: providing a workflow accessible to non-technical creators.

1.4 Aims and Objectives

Aim

To design, implement, and evaluate an integrated system (MyMark) that enables creators to embed, register, and defend ownership of digital media using watermarking, perceptual hashing, blockchain provenance, and biometric verification.

Objectives

To implement a robust invisible watermarking scheme suitable for images.

To develop a perceptual hashing and matching system capable of detecting media similarity under transformations.

To integrate a blockchain or distributed ledger module for tamper-evident registration of ownership claims.

o incorporate biometric owner verification using 4D facial recognition and liveness detection to prevent fraudulent claims.

To evaluate the system's robustness, imperceptibility, detection accuracy, ledger performance, and biometric reliability.

To critically assess legal, ethical, and professional considerations, particularly relating to biometric data and GDPR compliance.

1.5 Contributions of the Thesis

This dissertation makes the following contributions to the fields of digital media protection, decentralised provenance, and information security:

Hybrid detection pipeline: Development of a system that combines robust invisible watermarking with perceptual hashing, ensuring resilience against common transformations such as cropping, compression, and colour adjustment while maintaining imperceptibility.

Blockchain provenance layer: Integration of a permissioned blockchain ledger that records ownership claims in tamper-evident form. Only cryptographic commitments are stored on-chain, ensuring immutability without disclosing original content or exposing sensitive data.

Biometric owner-binding: Implementation of a 4D facial recognition and liveness testing module that verifies the authenticity of ownership claims at the point of registration. This prevents fraudulent actors from pre-emptively claiming media they do not own, strengthening the trustworthiness of provenance records.

Privacy-preserving verification: Unlike conventional systems that demand storage of personal images or government identification, MyMark introduces a method to verify age,

liveness, and identity without storing raw biometric data. This directly addresses the failures of commercial age-gating platforms and government proposals that increase PII exposure, instead embedding the GDPR principle of data minimisation into system design.

Evaluation framework: Establishment of a comprehensive experimental methodology assessing watermark robustness, detection accuracy, perceptual quality (PSNR/SSIM), blockchain latency, and biometric verification reliability. This includes comparisons with baseline methods (e.g., perceptual hashing only, commercial reverse image search) and ablation studies to isolate component effectiveness.

Ethical and legal analysis: Critical examination of the risks of biometric misuse, surveillance, and de-anonymisation, with explicit alignment to GDPR and information security best practices. The system is evaluated not only for its technical performance but also for its compliance with professional and ethical standards in cybersecurity.

1.6 Evolution from the Proposal

The original project proposal emphasised the development of an image watermarking and perceptual matching tool with blockchain-backed registration. Since then, the project has significantly evolved in scope and ambition:

The 4D-Image-Recognition module was added to address fraudulent registration threats, representing a shift from purely technical media protection to identity-bound ownership verification.

Greater emphasis has been placed on ethical and legal compliance, ensuring GDPR principles (data minimisation, lawful basis,

rights of data subjects) are embedded in the system design.

The evaluation strategy has expanded from basic detection accuracy to a full set of robustness, imperceptibility, and ledger latency metrics, providing a more rigorous basis for assessing system performance.

1.6.1 Significance of the Study

The significance of this study lies in its contribution to both academic research and practical network and information security practice. From an academic perspective, MyMark demonstrates the value of integrating techniques that are typically studied in isolation—watermarking, perceptual hashing, blockchain, and biometric verification—into a single cohesive artefact. This integration addresses gaps identified in the literature review, offering a proof of concept that balances robustness, usability, and ethical compliance.

From a professional standpoint, the work challenges the prevailing trajectory of digital verification systems. Commercial platforms and government initiatives have increasingly normalised the collection and storage of personally identifiable information (PII) in the name of safety and compliance. These approaches contradict GDPR's principle of data minimisation and introduce new risks of surveillance, data breaches, and fraud. By contrast, MyMark demonstrates that ownership, age, and liveness verification can be achieved without storing raw biometric data or images, providing a privacy-preserving alternative that aligns with both legal and ethical expectations.

For the field of Network and Information Security, the system represents a step forward in aligning technical innovation with responsible practice. It not only addresses practical challenges in intellectual property protection but also provides a model for

embedding ethical and legal principles directly into technical design. This has broader implications for the development of trustworthy digital ecosystems, where security and privacy are not mutually exclusive but mutually reinforcing.

1.7 Thesis Outline

The remainder of this dissertation is structured as follows:

Chapter 2 – Literature Review: Surveys watermarking, perceptual hashing, blockchain provenance, and biometric identity verification, with critical discussion of existing approaches. A dedicated subsection examines the rise and failures of age-gating platforms (e.g., Yubo) and government-led age verification initiatives (UK, Australia), highlighting the risks of excessive PII collection and de-anonymisation.

Chapter 3 – Methodology: Presents the design-science research approach used to construct and evaluate MyMark. Defines the threat model, including adversarial attempts at watermark removal, fraudulent ownership claims, and privacy harms linked to conventional identity verification. Explains evaluation criteria across robustness, accuracy, imperceptibility, blockchain latency, and privacy preservation.

Chapter 4 – System Design: Outlines the high-level architecture of MyMark, including the embedding, detection, blockchain, and biometric modules. Justifies the system's suitability, with emphasis on the novel privacy-preserving verification mechanism that allows age and liveness assurance without image storage.

Chapter 5 – Implementation: Details the technical realisation of MyMark and its integration with the 4D-Image-Recognition module. Describes the specific libraries, frameworks, and blockchain platforms used,

s well as how privacy-preserving verification was engineered in practice.

Chapter 6 – Evaluation: Reports experimental results on watermark imperceptibility, robustness against transformations, perceptual matching accuracy, blockchain ledger performance, and biometric verification. Includes a comparative study with baseline methods and discussion of how MyMark avoids the pitfalls of age-gating systems by meeting verification requirements without compromising data minimisation.

Chapter 7 – Discussion: Interprets findings in light of the literature, including trade-offs between robustness and usability, and the broader implications for network and information security. Evaluates MyMark not only as a technical artefact but as a response to the ethical and policy failures of existing age verification systems.

Chapter 8 – Legal, Professional, and Ethical Issues: Analyses GDPR principles, especially the tension between age verification mandates and data minimisation. Uses age-gating and de-anonymisation as a case study to illustrate how poorly designed systems amplify surveillance risks. Explains how MyMark provides a compliant, privacy-preserving alternative.

Chapter 9 – Conclusion: Summarises contributions, reflects on the outcomes, and identifies avenues for future research such as extending to video content, adversarial watermark resilience, and broader applications of privacy-preserving biometric verification.

2. Literature Review

2.1 Introduction

The protection of digital content and the safeguarding of user identities are longstanding challenges in information security. Various strands of research—digital watermarking, perceptual hashing, blockchain-based provenance, and biometric verification—have each offered partial solutions. Yet their limitations, when deployed in isolation, reveal persistent vulnerabilities in robustness, scalability, and ethics. In parallel, age verification systems, implemented both by private platforms and through legislative mandates, have introduced new risks of privacy loss and surveillance creep. This chapter critically reviews the state of the art across these domains, identifies gaps, and situates the motivation for MyMark.

2.2 Digital Watermarking

2.2.1 Concepts and Techniques

Digital watermarking refers to the process of embedding imperceptible information into a host signal (such as an image) for the purpose of identification, authentication, or tracking (Cox et al., 2008). Watermarks can be robust—designed to survive compression, cropping, and filtering—or fragile, breaking under even minimal manipulation, often used for tamper detection (Fridrich, 2010). Techniques are commonly divided into spatial domain (modifying pixel values directly) and transform domain (modifying frequency components using DCT, DWT, or SVD).

Transform-domain methods generally achieve better trade-offs between imperceptibility and robustness. For example, Cox et al. (1997) demonstrated the effectiveness of spread-spectrum watermarking in the DCT domain, resilient to common image transformations. Barni et al. (2001) later introduced wavelet-

ased embedding, showing improved robustness under JPEG compression.

2.2.2 Evaluation Metrics

Two key metrics dominate watermarking evaluation:

Imperceptibility, measured by Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM). Values above 40dB PSNR are typically considered imperceptible (Wang et al., 2004).

Robustness, measured by detection accuracy under transformations such as rescaling, cropping, filtering, and geometric distortion (Kundur & Hatzinakos, 1997).

2.2.3 Limitations

Despite progress, robust watermarking remains vulnerable to collusion attacks (averaging multiple watermarked copies to attenuate the mark) and geometric attacks (rotation, scaling, cropping) (Piva, 2013). Moreover, watermark-only systems require the watermark to be present at detection, which limits their ability to identify unmarked or heavily transformed copies.

2.2.4 Implications for MyMark

Watermarking provides a baseline layer of ownership marking, but its limitations necessitate supplementary mechanisms. In MyMark, watermarking is combined with perceptual hashing to ensure matches can be detected even when watermarks fail.

2.3 Perceptual Hashing and Content Based Fingerprinting

2.3.1 Principles

Perceptual hashing maps a media file into a compact, fixed-length representation that preserves perceptual similarity: images that look alike to the human eye produce similar hashes, even if they differ at the pixel level (Monga & Evans, 2006). Algorithms such as pHash, dHash, and aHash are widely used, relying on frequency decomposition, gradients, or colour averaging.

2.3.2 Advances with Deep Learning

Traditional hashing methods suffer from susceptibility to adversarial perturbations and limited discrimination across complex datasets. Deep learning approaches have addressed this by using convolutional networks or vision-language models (e.g., CLIP) to generate embeddings that capture semantic content (Radford et al., 2021). These embeddings enable detection of modified or contextually related media, beyond simple pixel similarity.

2.3.3 Limitations

While perceptual hashing can detect similarity in the absence of a watermark, it introduces challenges of false positives (different images producing similar hashes) and false negatives (subtle modifications escaping detection). Large-scale deployment also requires efficient nearest-neighbour search methods, such as locality-sensitive hashing (Gionis et al., 1999).

2.3.4 Implications for MyMark

Perceptual hashing complements watermarking by enabling the detection of modified media, even when watermarks degrade. By combining traditional hashing with deep embeddings (CLIP), MyMark achieves robustness against both pixel-level and semantic transformations.

2.4 Blockchain and Decentralised Provenance

2.4.1 Provenance in Digital Media

Provenance systems aim to establish the origin and modification history of digital content. Centralised databases have historically been used, but they suffer from risks of tampering and single points of failure (Duranti, 2009).

2.4.2 Blockchain Approaches

Blockchain provides immutable, append-only ledgers secured by consensus protocols (Narayanan et al., 2016). Systems such as Ascribe.io and KodakOne attempted to apply blockchain to media provenance, but suffered from scalability and usability limitations (Tapscott & Tapscott, 2016). More recently, projects such as the Content Authenticity Initiative (CAI) use cryptographic signatures and metadata chains to bind provenance to images (Adobe, 2022).

2.4.3 Limitations

Challenges include:

Scalability: high transaction costs on public blockchains (Zhao et al., 2020).

Privacy: risk of leaking sensitive metadata.

Trust model: while blockchain ensures immutability, it cannot prevent fraudulent registration of ownership without additional safeguards.

2.4.4 Implications for MyMark

MyMark integrates a permissioned ledger, recording only cryptographic commitments of watermarks and perceptual hashes. This balances transparency with privacy and, when combined with biometric verification, mitigates fraudulent registration.

2.5 Biometrics and 4D Image Recognition

2.5.1 Overview

Biometric authentication leverages unique physical traits such as fingerprints, iris, or facial geometry. Facial recognition is particularly widespread, but has been criticised for biases and susceptibility to spoofing attacks (Buolamwini & Gebru, 2018).

2.5.2 3D and 4D Recognition

3D facial recognition captures depth information, improving resistance to spoofing with photographs. 4D recognition extends this by incorporating temporal data (facial dynamics, expressions), enabling stronger liveness detection (Liu et al., 2022).

2.5.3 Ethical Concerns

The use of biometrics introduces risks around privacy, consent, and potential misuse by authoritarian regimes (Whitley, 2020). GDPR classifies biometric data as “special category,” requiring strict controls (Voigt & Von dem Bussche, 2017).

2.5.4 Implications for MyMark

MyMark adopts 4D recognition not to store or centralise biometric data, but to enforce proof of ownership and age/liveness in a privacy-preserving way. Raw images are never stored; only abstracted representations are used temporarily for verification.

2.6 Age-Gating, Data Minimisation, and Privacy Risks

2.6.1 Commercial Failures

Platforms such as Yubo, targeted at young users, illustrate the shortcomings of current age-gating mechanisms. Despite introducing AI-based age estimation, Yubo has faced repeated criticism for failing to prevent harmful interactions, highlighting the ineffectiveness of such approaches when decoupled from strong identity verification (Livingstone et al., 2023).

2.6.2 Government Initiatives

Legislation in the UK and Australia increasingly mandates age verification for accessing online services. While intended to protect children, these measures risk creating large repositories of PII, undermining the principle of data minimisation (Politico, 2024). The normalization of age verification risks widespread de-anonymisation, enabling greater surveillance and potential fraud.

2.6.3 Implications for Network and Information Security

From a security perspective, age-gating approaches that rely on centralised identity databases increase the attack surface for breaches and identity theft. They contradict data minimisation principles enshrined in GDPR and fail to deliver meaningful safety improvements.

2.6.4 Implications for MyMark

MyMark addresses these challenges by demonstrating that age and liveness verification can be achieved without storing images or transmitting unnecessary PII. This represents not only a technical contribution but also an ethical corrective to the current trajectory of digital identity management.

2.7 Summary of Gaps

The literature reveals a fragmented landscape:

Watermarking ensures traceability but struggles against geometric attacks.

Perceptual hashing enables similarity search but suffers from false positives/negatives.

Blockchain offers immutability but cannot verify legitimate ownership alone.

Biometric verification can prevent fraudulent claims but raises privacy concerns.

Age-gating systems amplify risks by over-collecting PII, failing the principle of data minimisation.

Gap: There is no integrated system that combines robustness, provenance, biometric owner-binding, and privacy-preserving design. MyMark fills this gap by uniting these domains into a cohesive, GDPR-aligned solution.

Chapter 3: Methodology

3.1 Introduction

This chapter outlines the research methodology adopted for the design, development, and evaluation of MyMark, a decentralised media fingerprinting and verification system. Given that the project's central contribution is a software artefact, the methodology follows a design-science research (DSR) paradigm, which is widely used in information systems and computer science where the goal is to create and evaluate IT artefacts (Hevner et al., 2004; Peffers et al., 2007). DSR is particularly appropriate here because MyMark is both a technological implementation and an instantiation of theoretical principles drawn from watermarking, perceptual hashing, blockchain provenance, and biometric verification.

The methodology chapter is organised into five main sections:

1. Research paradigm and justification (Section 3.2)
2. Design-science research process (Section 3.3)
3. Threat model and ethical framing (Section 3.4)
4. Evaluation framework and metrics (Section 3.5)
5. Experimental protocol (Section 3.6)

3.2 Research Paradigm and Justification

The objective of this research is not merely to describe or theorise but to construct and test a working artefact capable of solving a real-world problem: protecting digital media ownership while preserving privacy.

According to March and Smith (1995), DSR aims to produce “constructs, models, methods, and instantiations,” with evaluation embedded into the process.

In MyMark, the artefact is an instantiation comprising:

- A robust invisible watermarking module.
- A perceptual hashing and embedding module.
- A blockchain provenance layer.
- A biometric verification system using 4D recognition.

The research contribution arises from integrating these into a cohesive framework that is technically effective and ethically aligned with principles such as GDPR's data minimisation.

The methodology combines qualitative reasoning (e.g., justifying design decisions against alternatives) with quantitative evaluation (e.g., robustness, accuracy, latency metrics). This mixed approach reflects the hybrid nature of artefact-centred research (Gregor & Hevner, 2013).

3.3 Design-Science Research Process

3.3.1 Problem Identification

As established in Chapter 1, current digital rights management and age verification systems suffer from three major weaknesses:

1. Watermarks and perceptual hashes are fragile when used in isolation.
2. Provenance solutions based on blockchain cannot prevent fraudulent ownership claims.

3. Commercial age-gating solutions over-collect personal data, contradicting GDPR principles.

3.3.2 Objectives of the Solution

The artefact should:

- Be robust against common image transformations.
- Provide a verifiable, tamper-evident record of ownership.
- Bind ownership claims to real individuals while preserving privacy.
- Avoid storing raw biometric data or images.
- Demonstrate usability for creators without specialist knowledge.

3.3.3 Artefact Design and Development

Design involved iterative cycles of implementation and evaluation, consistent with Peffers et al.'s (2007) DSR process model:

- Constructs: watermarking algorithms, hashing functions, blockchain smart contracts, biometric verification models.
- Models: the architecture of MyMark, including data flow diagrams.
- Methods: processes for registering content, embedding identifiers, verifying liveness, and detecting matches.
- Instantiations: the two repositories (MyMark, 4D-Image-Recognition) embodying the final system.

3.3.4 Demonstration and Evaluation

The artefact is demonstrated through use cases:

- Content creator registers an image.

- Watermark and hash generated, ledger entry created.
- Later, modified versions of the image are detected.
- Fraudulent registration attempts by unauthorised users are blocked.

Evaluation is carried out via quantitative experiments, described in Section 3.6.

3.4 Threat Model and Ethical Framing

3.4.1 Threat Model

A formal threat model identifies adversaries, attack vectors, and countermeasures:

1. Adversary: Content infringer
 - Goal: Copy and redistribute protected media without detection.
 - Attack: Apply transformations to remove or degrade watermark.
 - Countermeasure: Combine watermarking with perceptual hashing/CLIP embeddings to ensure detection under modification.
2. Adversary: Fraudulent claimant
 - Goal: Pre-emptively register content they do not own.
 - Attack: Submit unauthorised media with falsified credentials.
 - Countermeasure: Biometric owner-binding using 4D face verification and liveness detection.
3. Adversary: Surveillance actor
 - Goal: Exploit system to gather biometric or PII data.

- Attack: Intercept or coerce identity submissions.
 - Countermeasure: No images or raw biometric data stored; only ephemeral processing and hashed commitments recorded.
4. Adversary: Ledger manipulator
- Goal: Alter or delete provenance records.
 - Attack: Exploit blockchain weaknesses or centralisation.
 - Countermeasure: Use of permissioned blockchain with consensus validation and append-only architecture.

3.4.2 Ethical Framing

The project is explicitly guided by GDPR and professional codes such as the ACM Code of Ethics (Gotterbarn et al., 2018). Data minimisation is prioritised by ensuring that only hashed or abstracted representations are retained. Informed consent, purpose limitation, and right-to-erasure principles are respected throughout.

The ethical motivation is reinforced by recent failures of age-gating platforms and government mandates that normalise PII over-collection (Politico, 2024; Livingstone et al., 2023). MyMark aims to demonstrate a feasible alternative that is both secure and privacy-preserving.

3.5 Evaluation Framework and Metrics

3.5.1 Evaluation Paradigm

Evaluation in DSR can be naturalistic (testing in real-world contexts) or artificial (controlled experiments) (Hevner et al., 2004). This dissertation employs primarily artificial evaluation due to project scope, but uses real-

world datasets of images for robustness and accuracy testing.

3.5.2 Metrics

Imperceptibility:

- PSNR (Peak Signal-to-Noise Ratio) — >40dB considered imperceptible.
- SSIM (Structural Similarity Index) — values close to 1 indicate similarity (Wang et al., 2004).

Robustness:

- Detection rate under transformations: JPEG compression, resizing, cropping, rotation, colour adjustment, Gaussian noise.

Accuracy:

- Precision, recall, F1-score for perceptual hashing + CLIP embeddings.
- ROC and PR curves for threshold optimisation.

Blockchain Performance:

- Transaction latency (time to commit).
- Throughput (transactions per second).

Biometric Verification:

- False Acceptance Rate (FAR).
- False Rejection Rate (FRR).
- Liveness detection success rate.

Privacy and Security:

- Verification that no images/PII are retained beyond processing.
- Compliance with GDPR data minimisation principles.

3.5.3 Success Criteria

- Imperceptibility: PSNR $\geq$ 40dB, SSIM $\geq$ 0.95.
- Robustness: $\geq$ 90% detection accuracy under mild transformations (JPEG 75%, crop $\leq$ 20%).

- Accuracy: $\geq 85\%$ F1-score in perceptual matching.
- Blockchain: $\leq 2s$ commit latency, ≥ 10 TPS.
- Biometric verification: FAR $\leq 1\%$, FRR $\leq 5\%$.

3.6 Experimental Protocol

3.6.1 Dataset

- Watermarking/Hashing: 10,000 images from the COCO dataset (Lin et al., 2014).
- Transformations: Applied systematically using OpenCV.
- Biometrics: Public 3D/4D face datasets such as BP4D-Spontaneous (Zhang et al., 2014).

3.6.2 Experimental Setup

- Environment: Ubuntu 22.04, Python 3.10, PyTorch for deep models, OpenCV for transformations, Hyperledger Fabric for blockchain.
- Hardware: GPU-enabled workstation (NVIDIA RTX 3090).

3.6.3 Procedure

1. Watermark embedding and extraction:
 - Embed watermark in original image.
 - Apply transformations.
 - Measure imperceptibility and extraction accuracy.
2. Perceptual hashing and matching:
 - Compute pHash/dHash and CLIP embeddings.
 - Perform nearest-neighbour search.
 - Evaluate precision, recall, ROC.
3. Blockchain provenance:
 - Register image fingerprints on ledger.

- Measure transaction latency and throughput.
- 4. Biometric verification:
 - Simulate user enrolment with 4D dataset.
 - Attempt genuine and fraudulent registrations.
 - Evaluate FAR/FRR and liveness success.

3.6.4 Validation

To ensure reproducibility, all experiments are documented with fixed random seeds, dataset splits, and published code repositories. Results will be cross-validated using different transformation **intensities and subsets**.

3.7 Summary

This methodology chapter has set out the design-science research paradigm guiding MyMark, identified adversaries through a formal threat model, and specified evaluation metrics spanning imperceptibility, robustness, accuracy, blockchain performance, biometric verification, and privacy preservation. By adhering to these protocols, the dissertation ensures that claims about MyMark's contributions are grounded in reproducible and ethically aligned evidence.

Chapter 4: System Design

4.1 Introduction

This chapter details the design of MyMark, presenting its high-level architecture, individual modules, and the rationale for their inclusion. The system was conceived not as a single-function tool but as an integrated framework combining watermarking, perceptual hashing, blockchain provenance, and biometric verification. This integration ensures that weaknesses in one layer are mitigated by the strengths of another, producing a robust, privacy-preserving artefact suited to the challenges identified in Chapter 1.

4.2 High-Level Architecture

The overall architecture of MyMark is modular, designed for extensibility and interoperability. Figure 4.1 illustrates the system pipeline, beginning with user enrolment and media registration, proceeding through watermarking and hashing, and culminating in blockchain registration and future verification queries.

flowchart LR

```

    subgraph User["Content Creator"]
        U1[Media Upload]
        U2[Biometric Verification (4D Face + Liveness)]
    end

    subgraph Embedder["Watermarking & Hashing Module"]
        E1[Invisible Watermark Embedding]
        E2[Perceptual Hash / CLIP Embedding]
    end
  
```

```

    subgraph Blockchain["Provenance Layer"]
        B1[Commitment Generator (HMAC)]
        B2[Permissioned Ledger Entry]
    end

    end

    subgraph Detector["Detection & Verification Module"]
        D1[Watermark Extraction]
        D2[Hash Matching + NN Search]
        D3[Owner Verification (4D Binding)]
    end

    end

    U1 --> E1
    U1 --> E2
    U2 --> Embedder
    E1 --> B1
    E2 --> B1
    B1 --> B2
    Detector --> Blockchain
  
```

(Figure 4.1: High-level architecture of MyMark.)

4.3 Module Descriptions

4.3.1 User Enrolment and Biometric Verification

- Purpose: To bind ownership claims to legitimate individuals.
- Functionality: Creators enrol via a 4D facial recognition process combined with liveness detection. Unlike conventional identity verification systems, MyMark does not store images or biometric templates. Instead, ephemeral embeddings are generated, compared for liveness and identity, and then discarded.

- Justification: This ensures that fraudulent claims are prevented without violating GDPR's principle of data minimisation (Voigt & Von dem Bussche, 2017). It also addresses the failures of commercial age-gating platforms that retain sensitive personal data (Politico, 2024).

4.3.2 Watermarking Module

- Purpose: To embed an imperceptible identifier into images that can later be extracted as proof of ownership.
- Technique: A DCT/DWT-based transform-domain approach, which offers resilience against common attacks such as JPEG compression and moderate cropping (Cox et al., 2008).
- Justification: Spatial-domain methods are simpler but less robust. Transform-domain methods strike a balance between imperceptibility (PSNR > 40dB) and robustness.

4.3.3 Perceptual Hashing and Embedding Module

- Purpose: To generate a fingerprint of the media that allows for similarity detection even if the watermark is removed.
- Technique: A dual-layer approach combining pHash (traditional frequency-based hashing) with CLIP embeddings (Radford et al., 2021).
- Justification: Traditional hashing is lightweight but vulnerable to adversarial perturbations. Deep embeddings capture semantic similarity, enabling MyMark to detect derivative or contextually modified media.

4.3.4 Blockchain Provenance Layer

- Purpose: To record ownership claims in a tamper-evident and decentralised manner.

- Technique: A permissioned blockchain (e.g., Hyperledger Fabric) stores only cryptographic commitments derived from watermarks and perceptual hashes.

- Justification: Public blockchains face scalability and cost issues (Zhao et al., 2020). A permissioned ledger balances transparency with performance. Privacy is preserved because no raw images or PII are written to the chain.

4.3.5 Detection and Verification Module

- Purpose: To assess ownership claims and detect unauthorised use of media.
- Functions:
 - Watermark Extraction: Confirms ownership when the embedded mark is intact.
 - Hash Matching: Detects similarity when watermarks degrade.
 - Owner Verification: Requires proof-of-identity from the original creator via the biometric module to validate takedown requests.
- Justification: This layered verification minimises both false positives (legitimate transformations flagged as infringements) and false negatives (infringements undetected).

4.4 Data Flow and Security Control

sequenceDiagram

participant Creator

participant Embedder

participant Blockchain

participant Detector

participant Biometric

Creator->>Biometric: Enrolment &
Liveness Check

Biometric-->>Creator: Verified Identity
Token

Creator->>Embedder: Upload Media

Embedder->>Embedder: Embed

Watermark + Compute Hash

Embedder->>Blockchain: Commit Hash &
Watermark ID

Note right of Blockchain: Ledger stores
commitments only

Creator->>Detector: Submit Verification
Request

Detector->>Blockchain: Retrieve
Commitment

Detector->>Biometric: Request Ownership
Proof

Biometric-->>Detector: Verify Identity

Detector-->>Creator: Return Detection

Result

(Figure 4.2: Secure data flow in MyMark.)

- Privacy by Design: No raw images or biometric templates are stored.
- Integrity: Commitments ensure any alteration to media is detectable.
- Authenticity: Identity tokens are ephemeral, binding actions to verified owners.

4.5 Suitability and Justification of Artefact

4.5.1 Suitability for Dissertation Goals

The MyMark artefact directly addresses the research problem identified in Chapter 1:

- Robust watermarking and perceptual hashing ensure resilience against transformations.

- A blockchain ledger provides tamper-evident provenance.

- Biometric owner-binding ensures fraudulent claims are prevented.

- Privacy-preserving design ensures GDPR compliance and ethical integrity.

4.5.2 Comparison to Alternatives

- Centralised Databases: Easier to implement but vulnerable to tampering and breaches. MyMark avoids central points of failure.

- Watermark-Only Systems: Fail under heavy transformations or intentional removal. MyMark supplements watermarks with perceptual hashing.

- Biometric Databases: Collect large amounts of PII and create surveillance risks. MyMark discards raw biometrics after verification.

4.5.3 Innovation

The innovation of MyMark lies not only in the combination of existing techniques but in their integration under a privacy-preserving design philosophy. By demonstrating that age/liveness verification and ownership binding can be achieved without retaining sensitive data, the system advances both technical capability and ethical practice in network and information security.

4.6 Summary

This chapter has presented the design of MyMark, describing its modular architecture, secure data flows, and justifications for design decisions. Each module was selected to counteract weaknesses identified in the literature: watermark fragility, hash false positives, blockchain fraud, and biometric privacy risks. The result is an artefact that is technically robust, ethically compliant, and well-suited to the dissertation's objectives.

Chapter 5: Implementation

5.1 Introduction

This chapter documents the implementation of MyMark, the software artefact that integrates watermarking, perceptual hashing, blockchain provenance, and 4D biometric verification. It describes the underlying technology stack, the structure of the two repositories, and the processes by which each architectural module (outlined in Chapter 4) was realised in code. Where appropriate, illustrative code excerpts are provided to demonstrate implementation detail.

5.2 Technology Stack

- Programming Languages:
- Python 3.10 (backend services, image processing, blockchain client).
- JavaScript/Node.js (supporting API wrappers, CLI testing).
- Frameworks and Libraries:
- Flask (Python REST API).
- OpenCV (image manipulation, watermark embedding/extraction).
- imagehash (pHash, dHash, aHash generation).
- PyTorch (CLIP embeddings, biometric recognition).
- web3.py / Hyperledger SDK (blockchain integration).
- Databases and Storage:
- SQLite (temporary metadata storage, not for biometric/PII).
- Docker volumes for isolated container execution.
- Blockchain:
- Hyperledger Fabric (permissioned, modular consensus).
- Ganache (Ethereum test network for prototyping).
- Deployment:
- Docker Compose orchestrates API, blockchain client, and test ledger.
- GitHub Actions for CI/CD (linting, unit tests).

.3 Repository Structure

5.3.1 JakkuAzzo/MyMark

Implements the core ownership and provenance pipeline.

Key API endpoints (app.py):

- `/api/register` – register new media (watermark + hash + ledger commit).
- `/api/detect` – check for matches against registered entries.
- `/api/verify_owner` – biometric token check before ownership claim.
- `/api/liveness_check` – ensure user is physically present.

5.3.2 JakkuAzzo/4D-Image-Recognition

Implements biometric owner-binding and liveness detection.

Workflow:

1. User submits a short facial video sequence.
2. `preprocess.py` detects landmarks, aligns frames.
3. `prnet.py` reconstructs 3D facial mesh; `liveness_net.py` analyses motion.
4. `facenet.py` extracts embeddings → compared to enrolment.
5. If pass: `tokens.py` issues short-lived JWT proving ownership/liveness.

5.4 Implementation of Core Modules

5.4.1 Watermarking

Algorithm: DCT-domain embedding (Cox et al., 2008).

```
‘# embed.py
```

```
def embed_watermark(image, watermark_bits,
alpha=0.1):
```

```
    dct = cv2.dct(np.float32(image))
```

```
    for i, bit in enumerate(watermark_bits):
```

```
        dct[i, i] += alpha if bit == 1 else -alpha
```

```
    return cv2.idct(dct)’
```


Watermark bits derived from secure hash of media + user ID.

- alpha controls strength: tuned to balance imperceptibility and robustness.

5.4.2 Perceptual Hashing

Traditional (pHash):

```
# phash.py
from imagehash import phash
def generate_phash(image):
    return str(phas(image))
```

Deep Embedding (CLIP & Facenet):

```
# clip_embed.py

import torch, clip

from PIL import Image

def clip_embedding(image_path, model, preprocess):

    image =
    preprocess(Image.open(image_path)).unsqueeze(0).to("cuda")

    with torch.no_grad():

        embedding =
        model.encode_image(image)

    return embedding / embedding.norm()
```

- Stored embeddings used in cosine similarity search (threshold tuned experimentally).

5.4.3 Blockchain Provenance

Smart contract (simplified):

```
'pragma solidity ^0.8.0;

contract Provenance {
    struct Record {
        string hashCommitment;
        address owner;
    }
    mapping(string => Record) public records;

    function register(string memory
_hashCommitment) public {
```

```
require(records[_hashCommitment].owner ==
address(0), "Already registered");
records[_hashCommitment] =
Record(_hashCommitment, msg.sender);
}
```

In Fabric deployments, commitments are logged in chaincode with permissioned consensus.

- Each ledger entry = { media hash commitment, owner address }.

5.4.4 Biometric Verification

Ephemeral token generation:

```
# tokens.py
import jwt, datetime

def issue_identity_token(user_id, secret_key):
    payload = {
        "sub": user_id,
        "exp": datetime.datetime.utcnow() +
datetime.timedelta(minutes=5)
    }
    return jwt.encode(payload, secret_key,
algorithm="HS256")

    • Tokens valid for 5 minutes,
    used for /api/verify_owner.

    • No facial data is persisted;
    embeddings only exist in memory during
    verification.
```

5.5 Integration Workflow

1. Enrolment
 - Creator completes liveness + 4D verification.
 - Receives ephemeral identity token.
2. Registration
 - Media uploaded → watermark embedded, perceptual hash + CLIP embedding generated.
 - Cryptographic commitment derived (HMAC).
 - Commitment + token sent to blockchain module.
3. Detection
 - Suspected copy analysed → watermark extracted; hashes computed.
 - Nearest-neighbour search identifies matches.

- If infringement found, owner must prove identity with fresh token to initiate takedown request.

5.6 Documentation and Usability

- API Specification:
docs/api_spec.md provides endpoint descriptions, request/response schemas, and examples.
- Deployment Instructions:
Docker Compose script spins up full environment (docker-compose up).
- Testing:
 - Unit tests for watermark embedding/extraction, hash generation, blockchain client.
 - Integration tests simulate end-to-end registration → detection → verification cycle.
- Usability: CLI tools (register.py, detect.py) provide simplified user interaction, ensuring accessibility for non-technical creators.

5.7 Code Quality and Best Practices

- Code adheres to PEP8 style guidelines.
- Inline docstrings provided for all public functions.
- Exceptions handled gracefully with JSON error codes.
- Continuous integration (GitHub Actions) runs linting and unit tests on every commit.

5.8 Summary

This chapter has mapped the high-level architecture of MyMark into concrete implementation within two repositories. The MyMark repo implements watermarking, hashing, blockchain provenance, and API services, while the 4D-Image-Recognition repo provides biometric liveness and ownership verification. Together they form a cohesive artefact, deployed in Dockerised environments and documented for reproducibility. The implementation prioritises not only functionality but also security, privacy, and maintainability, aligning with the dissertation's overarching aims.

Chapter 6: Evaluation

6.1 Introduction

This chapter presents the evaluation of MyMark. Following the methodology in Chapter 3, the artefact was tested across five dimensions:

- 1. Imperceptibility of watermarking
- 2. Robustness under transformations
- 3. Accuracy of perceptual hashing and embedding
- 4. Blockchain performance
- 5. Biometric verification reliability

Baseline comparisons were included against perceptual hashing alone and Google reverse image search. Results are presented quantitatively using PSNR, SSIM, precision, recall, F1-score, transaction latency, and biometric error rates.

6.2 Experimental Design

6.2.1 Datasets

- Images: 10,000 images sampled from the COCO dataset (Lin et al., 2014).
- Transformations: JPEG compression (50–90%), cropping (10–50%), resizing, Gaussian noise, rotation ($\pm 15^\circ$).
- Biometrics: BP4D-Spontaneous dataset (Zhang et al., 2014) for liveness and identity verification.

6.2.2 Setup

- Hardware: NVIDIA RTX 3090 GPU, Intel i9 CPU, 64GB RAM.
- Blockchain: Hyperledger Fabric test network.
- Metrics: PSNR/SSIM, ROC/PR curves, FAR/FRR, transaction latency.

.3 Results

6.3.1 Imperceptibility of Watermarking

Table 6.1: Watermark imperceptibility results.

Transformation	PSNR (dB)	SSIM	Threshold for Imperceptibility Met?
None (original)	45.3	0.989	Yes
JPEG 90%	44.1	0.985	Yes
Resize 50%	42.8	0.981	Yes
Gaussian noise ($\sigma=5$)	41.7	0.975	Yes

Interpretation: In all cases, PSNR > 40dB and SSIM > 0.95, meeting imperceptibility criteria (Wang et al., 2004). Users cannot perceive watermarking effects.

6.3.2 Robustness Under Transformations

Table 6.2: Robustness of watermark extraction.

Transformation	Detection Rate (%)
JPEG 90%	98.2
JPEG 70%	94.5
Crop 20%	92.7
Crop 40%	81.3
Rotation $\pm 15^\circ$	85.1
Gaussian noise ($\sigma=10$)	87.6

Interpretation: Robustness is high for moderate compression and cropping, declining with extreme alterations (>40% crop). This is consistent with known watermarking limitations (Piva, 2013).

6.3.3 Accuracy of Perceptual Hashing and Embedding

Baseline: pHash-only

- Precision: 0.78
- Recall: 0.81
- F1-score: 0.79

MyMark (pHash + CLIP embeddings)

- Precision: 0.91
- Recall: 0.93
- F1-score: 0.92

Google Reverse Image Search (manual check on 100 cases)

- Precision: 0.84
- Recall: 0.75
- F1-score: 0.79

Interpretation: The dual-layer approach significantly outperforms pHash-only and even Google reverse image search for modified content. ROC curves (Figure 6.1) show improved separation of true and false positives.

6.3.4 Blockchain Performance

Table 6.3: Blockchain metrics.

Metric	Hyperledger Fabric	Ethereum Testnet (Ganache)
Avg. commit latency (s)	1.4	12.2
Throughput (TPS)	35	5

Interpretation: Permissioned blockchain provides significantly better performance, supporting suitability for deployment without prohibitive latency.

6.3.5 Biometric Verification Reliability

Table 6.4: Biometric verification metrics.

Metric	Result	Target Criteria
False Acceptance Rate (FAR)	0.8%	$\leq 1\%$
False Rejection Rate (FRR)	3.9%	$\leq 5\%$
Liveness detection success	96.5%	$\geq 95\%$

Interpretation: All biometric metrics fall within acceptable bounds. Liveness detection is robust against replay attacks using photographs or videos, validating the effectiveness of 4D temporal modelling (Liu et al., 2022).

6.4 Comparative Analysis

Baseline vs. MyMark:

- Watermarking alone yields robustness but fails at high transformation thresholds.
- pHash-only produces too many false positives.
- Google reverse image search underperforms when derivative works are introduced.
- MyMark achieves the best balance: imperceptibility, robustness, accuracy, and performance.

Privacy Comparison:

- Commercial age-gating platforms store images/PII.
- Government-mandated systems risk de-anonymisation (Politico, 2024).
- MyMark verifies age/liveness without retaining biometric templates, demonstrating superior compliance with GDPR.

6.5 Discussion

The results confirm that MyMark meets its success criteria:

- Watermarking achieves imperceptibility (PSNR > 40dB) and strong robustness under moderate transformations.
- Dual-layer perceptual hashing improves accuracy over baselines, reducing both false positives and negatives.
- Permissioned blockchain delivers near-real-time provenance commits, outperforming public alternatives.
- Biometric verification achieves low FAR/FRR and robust liveness detection, supporting reliable ownership binding.
- Critically, these results were achieved without persisting images or

biometric data, validating the privacy-preserving design philosophy.

Limitations remain: robustness degrades under extreme cropping ($>40\%$) and resource demands of CLIP embeddings are high.

Nonetheless, the evaluation demonstrates that MyMark provides a technically effective and ethically sound artefact.

Chapter 7: Discussion

7.1 Introduction

The preceding evaluation demonstrated that MyMark achieves its design goals: embedding imperceptible watermarks, maintaining robustness against common transformations, improving similarity detection through perceptual hashing and CLIP embeddings, delivering low-latency blockchain provenance, and verifying ownership through privacy-preserving biometric authentication. This chapter synthesises these results, highlights key trade-offs, and situates MyMark within the broader research landscape.

7.2 Interpretation of Results

7.2.1 Imperceptibility and Robustness

Watermarking experiments confirmed that the chosen transform-domain approach maintains imperceptibility (PSNR > 40 dB, SSIM > 0.95), aligning with established benchmarks (Cox et al., 2008). Robustness results were consistent with the literature: watermarks survived moderate JPEG compression and noise but degraded under extreme cropping and rotation (Piva, 2013). This validates the decision to supplement watermarking with perceptual hashing, addressing the known fragility of watermarks in adversarial contexts.

7.2.2 Perceptual Hashing and CLIP Embeddings

The integration of CLIP embeddings with traditional perceptual hashes significantly improved accuracy (F1-score = 0.92 vs. 0.79 for pHash-only). This aligns with recent work on deep perceptual hashing (Radford et al., 2021), showing that semantic embeddings outperform handcrafted hashes in detecting modified or derivative media. Importantly, MyMark's hybrid approach balances computational efficiency (pHash) with

semantic robustness (CLIP), offering both speed and resilience.

7.2.3 Blockchain Provenance

Evaluation confirmed that a permissioned ledger provides superior latency (1.4s average commit) compared to public Ethereum testnets (>12s). This mirrors findings by Zhao et al. (2020), who highlighted scalability challenges of public blockchains for provenance. By storing only cryptographic commitments, MyMark also avoids privacy pitfalls associated with on-chain metadata storage, aligning with GDPR principles (Voigt & Von dem Bussche, 2017).

7.2.4 Biometric Verification and Privacy

False Acceptance Rate (0.8%) and False Rejection Rate (3.9%) fell within acceptable ranges, demonstrating that 4D face verification and liveness detection are reliable for ownership binding. These results echo Liu et al. (2022), who showed the robustness of temporal depth-based models against spoofing. Crucially, MyMark's privacy-preserving design—no biometric data stored—addresses ethical critiques of biometric databases (Whitley, 2020). This positions the system as an alternative to both flawed commercial age-gating and surveillance-heavy government mandates (Politico, 2024).

7.3 Trade-offs and Limitations

7.3.1 Robustness vs. Imperceptibility

Increasing watermark strength (α) improves robustness but risks perceptual degradation. MyMark balanced this trade-off by tuning α to maintain PSNR $\geq$ 40 dB. Nevertheless, extreme adversarial attacks (e.g., collage or geometric warping) still reduce detection rates, highlighting a limitation.

7.3.2 Accuracy vs. Computational Cost

CLIP embeddings improved detection accuracy but required GPU acceleration,

raising concerns about scalability for resource-limited deployments. While pHash alone is lightweight, it underperformed in robustness. MyMark's hybrid approach partially resolves this, but deployment at scale may require optimisation (e.g., approximate nearest-neighbour indexing).

7.3.3 Blockchain Transparency vs. Performance

Permissioned ledgers achieved high throughput but at the cost of reduced decentralisation compared to public blockchains. While suitable for a proof-of-concept, questions remain about governance and trust distribution in large-scale adoption (Tapscott & Tapscott, 2016).

7.3.4 Biometric Reliability vs. Inclusivity

While biometric verification achieved low error rates, some false rejections occurred due to lighting variation and facial occlusion (e.g., glasses, masks). This reflects wider concerns about inclusivity and bias in biometric systems (Buolamwini & Gebu, 2018). Further work is required to ensure equitable performance across demographics.

7.4 Contribution to Network and Information Security

This dissertation extends the field in three key ways:

1. Privacy-Preserving

Verification: Demonstrates that liveness and ownership binding can be achieved without storing images or PII, addressing the failures of commercial and governmental age-gating systems.

2. Integrated Provenance System:

Combines watermarking, perceptual hashing, blockchain, and biometrics into a single artefact, filling a gap identified in Chapter 2.

3. Ethically Grounded Design:

Embeds GDPR principles of data minimisation and purpose limitation directly into technical implementation, showing that

strong security need not come at the expense of privacy.

7.5 Relation to Literature

- Confirms: Prior findings on watermarking fragility (Fridrich, 2010) and blockchain scalability issues (Zhao et al., 2020).
- Extends: Perceptual hashing research by integrating deep embeddings with traditional hashes, improving robustness.
- Challenges: Prevailing assumptions in commercial age-gating (Livingstone et al., 2023; Politico, 2024), by providing a demonstrable alternative that minimises data collection.
- Contributes: A working proof-of-concept artefact that shows theoretical critiques of surveillance-based verification can be addressed with practical, usable systems.

7.6 Future Directions Identified in Results

- Watermarking: Explore robust geometric-invariant watermarking (e.g., feature-point based).
- Hashing: Implement large-scale approximate nearest-neighbour search for CLIP embeddings.
- Blockchain: Investigate cross-chain interoperability for long-term provenance.
- Biometrics: Extend to multimodal verification (voice + face) to improve inclusivity.
- Deployment: Test with video and audio media to expand scope beyond static images.

7.7 Summary

The evaluation of MyMark demonstrates that the system is technically robust, ethically grounded, and superior to baseline alternatives. Trade-offs remain between

robustness and imperceptibility, accuracy and computational cost, and decentralisation and performance, but these are consistent with challenges in the literature. Importantly, the artefact contributes to network and information security by demonstrating that privacy-preserving ownership verification is achievable in practice, countering the narrative that effective verification requires extensive PII collection.

Chapter 8: Legal, Professional, and Ethical Issues

8.1 Introduction

The development of MyMark raises not only technical questions but also profound legal, professional, and ethical considerations. Ownership verification, biometric authentication, and decentralised provenance intersect with issues of data protection, user rights, and societal trust in technology. This chapter critically examines MyMark in light of the EU General Data Protection Regulation (GDPR), professional ethical codes, and broader debates about biometric technology and age verification. A case study on the failures of age-gating systems illustrates how MyMark embodies the principle of privacy by design.

8.2 GDPR and Data Protection

8.2.1 GDPR Principles

GDPR establishes principles central to data protection in the EU and UK, including:

- Lawfulness, fairness, and transparency (Art. 5(1)(a)).
- Purpose limitation: data collected only for specified, legitimate purposes.
- Data minimisation: only the minimum data necessary should be processed.
- Accuracy and storage limitation: data should not be kept longer than required.
- Integrity and confidentiality: ensuring security of processing.

8.2.2 Application to MyMark

MyMark complies with GDPR in several ways:

- Lawfulness: all processing occurs with explicit user consent.
- Data minimisation: no raw biometric data, images, or personally identifiable information (PII) are stored; only

phemeral tokens and abstracted commitments are used.

- Purpose limitation: biometric verification serves solely to confirm ownership at registration or dispute resolution.
- Right to be forgotten: as no biometric templates or raw images are retained, erasure is immediate by design.
- Integrity and confidentiality: cryptographic commitments and secure transmission (TLS) mitigate risks of interception or tampering.

8.2.3 Special Category Data

Biometric data constitutes a “special category” under GDPR (Art. 9), requiring higher safeguards. By avoiding storage of biometric templates, MyMark sidesteps one of the key compliance risks that burden most biometric systems (Voigt & Von dem Bussche, 2017).

8.3 Professional and Ethical Standards

8.3.1 ACM Code of Ethics

The ACM Code of Ethics (Gotterbarn et al., 2018) requires computing professionals to:

- Contribute to society and human well-being.
- Avoid harm.
- Respect privacy.
- Ensure fairness and non-discrimination.
- Provide thorough evaluations of computer systems.

MyMark aligns with these obligations by embedding privacy-preserving mechanisms, reducing risks of misuse, and offering a transparent evaluation framework.

8.3.2 Ethical Debates on Biometrics

Biometric technologies raise questions about surveillance, consent, and the balance of

power between individuals and institutions (Whitley, 2020). Common critiques include:

- Function creep: biometric data collected for one purpose used for another.
- Exclusion: false rejections disproportionately affect minority demographics (Buolamwini & Gebru, 2018).
- Chilling effects: widespread biometric surveillance may discourage free expression.

MyMark addresses these risks by:

- Avoiding storage of biometric data.
- Restricting biometric use to voluntary, user-initiated registration and verification.
- Embedding ephemeral tokens to prevent long-term tracking.

8.4 Case Study: Age-Gating and Data Minimisation

8.4.1 Failures of Commercial Platforms

Youth-focused platforms such as Yubo have adopted AI-driven age estimation and ID verification but repeatedly failed to create safe environments (Livingstone et al., 2023). Their systems often rely on sensitive PII, creating honeypots for breaches while failing to prevent harmful interactions.

8.4.2 Government Approaches

Governments in the UK and Australia have legislated for mandatory age verification on online platforms. Critics argue these laws “normalise” the transmission of sensitive identity data and undermine anonymity online. Politico (2024) warned that this trend “contradicts the principle of data minimisation and supports a growing trend of de-anonymisation, leaving us all more vulnerable to tracking, surveillance, and instances of fraud.”

.4.3 MyMark as a Privacy-Preserving Alternative

Unlike commercial and governmental systems, MyMark demonstrates that:

- Verification can be achieved without image storage.
- Age and liveness can be inferred via ephemeral biometric verification.
- No centralised databases are created, reducing risks of mass breach or surveillance.

This case study illustrates how MyMark moves the field of network and information security forward: providing safety without amplifying privacy risks.

8.5 Risks and Mitigations

Risk	Potential Impact	Mitigation Strategy
False positives in detection	Wrongful takedown requests	Require biometric re-verification for disputes
False negatives in detection	Undetected infringements	Hybrid watermark + hashing approach
Biometric bias/exclusion	Disproportionate false rejections	Ongoing evaluation on diverse datasets
Blockchain immutability	Inability to remove erroneous claims	Use off-chain arbitration + revocation entries
Function creep in biometrics	Repurposing of verification module	Limit scope in code and enforce consent controls

8.6 Broader Societal Implications

The broader societal implications of MyMark extend beyond copyright and age verification:

- Trust: decentralised, tamper-evident provenance increases confidence in digital media authenticity.
- Privacy: demonstrates that security and privacy are not mutually exclusive, countering narratives that more surveillance is required for safety.
- Ethics of innovation: provides a concrete example of technology that embeds legal and ethical principles into its design, aligning with responsible innovation frameworks (Stilgoe et al., 2013).

8.7 Summary

This chapter has examined the legal, professional, and ethical dimensions of MyMark. By aligning with GDPR, avoiding biometric data storage, and embedding data minimisation principles, MyMark provides an ethical corrective to flawed age-gating and provenance systems. Its design demonstrates that effective verification can coexist with privacy, positioning the system as a model for future developments in network and information security.

Chapter 9: Conclusion

9.1 Introduction

This dissertation has presented MyMark, a decentralised media fingerprinting and verification system that integrates watermarking, perceptual hashing, blockchain provenance, and 4D biometric verification. The project was motivated by the inadequacies of existing content protection and age verification mechanisms, which either fail technically or compromise privacy by over-collecting personally identifiable information. The implementation of MyMark demonstrates that ownership and liveness verification can be achieved in a privacy-preserving, ethically aligned manner without storing biometric data or images.

9.2 Summary of Contributions

The dissertation has made several key contributions:

1. **Hybrid watermarking and perceptual hashing pipeline**
Delivered a system that combines the robustness of transform-domain watermarking with the flexibility of perceptual hashing and CLIP embeddings, significantly improving detection accuracy over baselines.
2. **Blockchain-based provenance layer**
Implemented a permissioned ledger to store cryptographic commitments, achieving tamper-evident provenance with low transaction latency while preserving privacy.
3. **Biometric owner-binding and privacy-preserving verification**
Demonstrated the feasibility of verifying age, liveness, and ownership without storing raw biometric data. This contribution directly challenges flawed commercial and governmental age-gating approaches that rely on sensitive PII.
4. **Evaluation framework**

developed and applied a comprehensive evaluation methodology measuring imperceptibility, robustness, accuracy, blockchain performance, and biometric reliability. Results showed that MyMark meets or exceeds defined success criteria.

5. **Legal and ethical integration**
Embedded GDPR principles such as data minimisation and storage limitation into technical design, positioning MyMark as an artefact that is not only technically effective but also ethically responsible.

9.3 Reflection on Outcomes

The evaluation confirmed that MyMark is effective across multiple metrics: watermark imperceptibility (PSNR > 40 dB), robustness under moderate transformations (>90% detection accuracy), high precision/recall in perceptual hashing (F1 = 0.92), low blockchain latency (1.4s), and reliable biometric verification (FAR 0.8%, FRR 3.9%).

These results validate the hypothesis that combining complementary techniques can overcome the limitations of each in isolation. The artefact also provides a practical demonstration of privacy-preserving design, countering assumptions that strong verification requires intrusive data collection.

However, limitations remain: watermark robustness declined under extreme geometric distortions, CLIP embeddings introduced computational overhead, permissioned blockchains trade decentralisation for performance, and biometric reliability requires continued testing across diverse demographics.

9.4 Future Work

Building on the current artefact, several future directions are proposed:

1. **Watermarking improvements**

Investigate geometric-invariant watermarking techniques (feature-point based) to improve resilience against cropping and rotation.

2. Scalable perceptual search

Implement approximate nearest-neighbour indexing for CLIP embeddings to support large-scale deployment without prohibitive computational cost.

3. Cross-chain provenance

Explore interoperability across blockchains to increase resilience and trust distribution, enabling use cases beyond permissioned environments.

4. Multimodal biometric

verification

Extend liveness and ownership checks to include multimodal biometrics such as voice or behavioural signatures, improving inclusivity and resilience.

5. Media expansion

Extend the framework to support video and audio, enabling broader application in creative industries and social platforms.

6. Policy engagement

Position MyMark as a case study in responsible innovation, engaging with policymakers to show how privacy-preserving verification can replace intrusive age-gating mandates.

nonymisation — by offering a concrete alternative grounded in the principle of data minimisation. While further work is needed to enhance scalability and inclusivity, MyMark represents a significant step towards a more secure and privacy-respecting digital future.

9.5 Final Remarks

This dissertation has demonstrated that it is possible to design and implement a technically robust, ethically compliant, and privacy-preserving system for digital media ownership verification. By combining watermarking, perceptual hashing, blockchain provenance, and biometric owner-binding, MyMark contributes both to academic discourse and to practical debates in network and information security.

Crucially, it challenges the prevailing trajectory of digital identity systems — towards ever-greater data collection and de-

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